

ASHRITH SAMBARAJU

Hyderabad, Telangana, 500097 | Phone: +91 9100673257 | E-Mail: ashrithsambaraju@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

SPHOORTHY ENGINEERING COLLEGE (JNTUH) - 7.93 / 10.0

Bachelors of Technology in Computer Science - AIML

Hyderabad, TG, INDIA

2021-2025

PUBLICATIONS

Lead Author - “Automated Brain Tumor Classification Using Hybrid Deep Learning Models”, published in the *International Research Journal of Advanced Engineering Hub (IRJAEH)*. DOI: [10.47392/IRJAEH.2025.0318](https://doi.org/10.47392/IRJAEH.2025.0318)

EXPERIENCE

Bharat Smart Services (Coral Innovations Pvt Ltd) - Data Science Intern

September 2025 - Present

- Developing an **AI-driven clustering framework** that integrates feature-based analytics with upcoming image-based modeling through **YOLO** and **OpenCV** to enhance meter recognition and automate classification accuracy
- Built an **AI-driven electricity theft detection system** using 5-second smart-meter data, combining **hybrid anomaly modeling**, spike/dip event analytics, and **NILM**-style appliance pattern estimation. Built a full end-to-end pipeline with **Isolation Forest**, **LSTM Autoencoder**, rule-based cue extraction, and an interactive **Streamlit dashboard** for real-time monitoring, alerts, and forensic analysis.
- Built a Streamlit-based data management dashboard with **MongoDB GridFS** for storing meter records, image handling, and **automated PDF reports**.

NextGen Edunet Foundation X EY - MERN Intern

March 2025 - April 2025

- Built a dynamic **e-commerce web application** using the **MERN** stack as part of the Internship under Edunet Foundation in collaboration with **EY**, focusing on seamless user experience, user authentication, and efficient product management.
- Structured the application around a **RESTful API**, ensuring the React frontend communicated efficiently and reliably with the MongoDB backend.
- Applied knowledge of secure practices** by integrating **JWT (JSON Web Tokens)** for robust user authentication and authorization, protecting session data.

International Institute of Information Technology (IIIT-H X Swecha) - AI Developer Intern

May 2024 - June 2024

- Completed a five-day intensive program focusing on AI, **Large Language Models (LLMs)**, system architecture, and building custom operating systems (Debian), leveraging open-source technology.
- Explored the application of **AI**, particularly LLMs, in real-world contexts, specifically engaging in discussions on AI-driven solutions for preserving **Telugu culture**.
- Received mentorship from **IIT & IIIT alumni** on portfolio building and career strategies, while fostering collaborative problem-solving skills through cross-institutional knowledge sharing.

PROJECTS

- AI-Powered Brain Tumor Detection Treatment Recommendation System:** [Link](#)

Built a hybrid deep learning model achieving **85% Accuracy** using **VGG16**, **ResNet-50**, and **EfficientNetB2** with **Grad-CAM** for automated brain tumor detection. Integrated treatment suggestions, symptom-based screening, and PDF report generation to support clinical decisions.

- E-Commerce Web Application (MERN Stack):** [Link](#)

Developed full-stack e-commerce web application using the **MERN** stack with product listing, authentication, and cart features. Built responsive interfaces with React, implemented **REST APIs**, and managed data handling with **MongoDB**.

- Blockchain-Based Secure Academic Credential Management System:** [Link](#)

Designed a secure blockchain system using **Ethers** and **OTP authentication** for tamper-proof credential issuance, storage, and verification. Ensures transparency and integrity for universities, students, and verifiers

SKILLS

Programming Languages

- Python Programming, R.

AI&ML Frameworks

- NumPy, Pandas, Matplotlib, SciKit-Learn, YOLO, OpenCV, LLM's, PyTorch, Streamlit.

Database & Data Storage

- Oracle SQL, MongoDB.

IDEs

- Visual Studio, Jupyter, Google Colab

Frontend Development

- HTML, CSS, React, Bootstrap.

Soft Skills

- Communication, Networking, Problem Solving, Prioritization & Task Management.